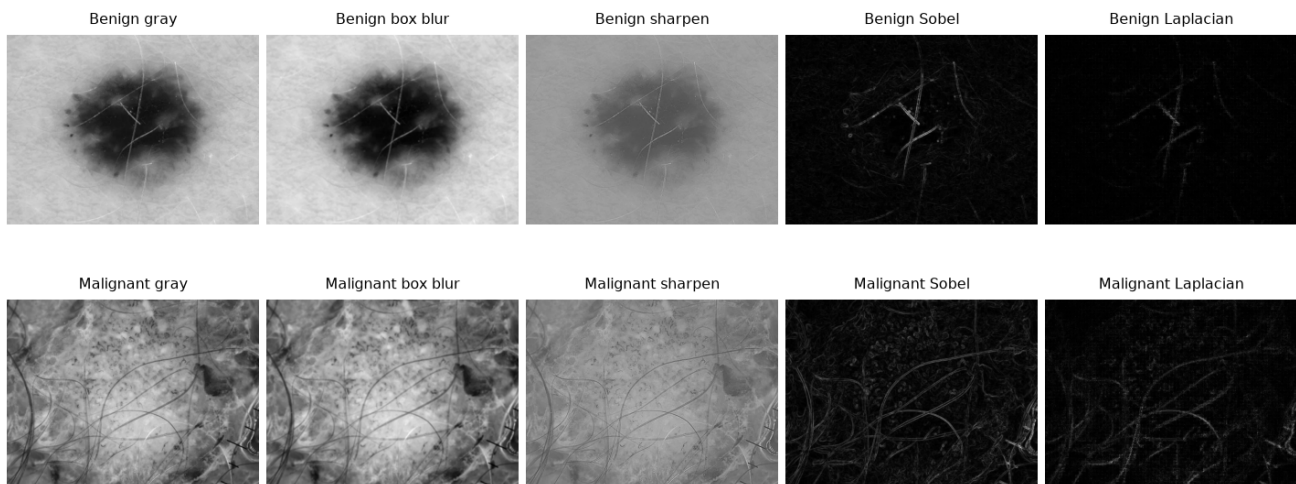


Filters on MILK10k images

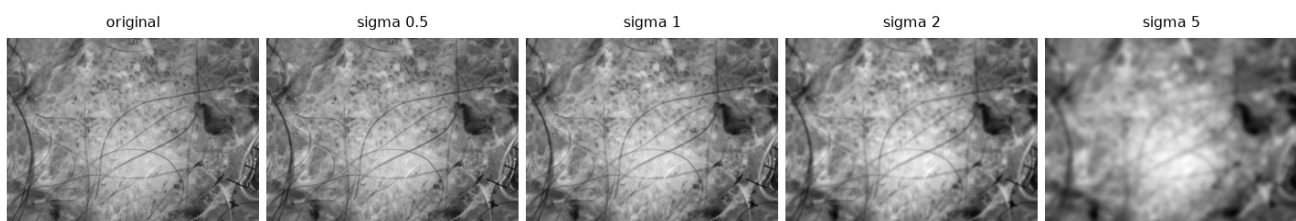
Q1. Do edges or textures look different between malignant and benign lesions?

Benign ISIC_0073863: mean Sobel 16.4, mean absolute Laplacian 3.2. Malignant ISIC_0051817: mean Sobel 42.3, mean absolute Laplacian 7.2. On this pair the malignant photo has stronger edges (42.3 vs 16.4) and stronger texture (7.2 vs 3.2). The kernels make structure easier to see, but one pair is not a class rule. Lighting, scale, and how much of the frame is lesion also change both numbers.



Q2. What does sigma control in gaussian_filter?

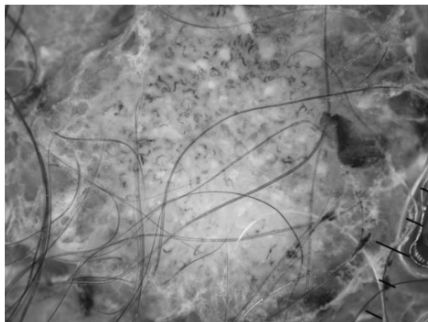
In `scipy.ndimage.gaussian_filter`, sigma is the standard deviation of the Gaussian kernel, measured in pixels. A small sigma uses a narrow kernel and only mixes each pixel with its close neighbours. A large sigma uses a wide kernel, so fine edges and texture are averaged away. The default kernel radius is $\text{round}(4 * \text{sigma})$. On the malignant photo the mean Sobel falls as sigma grows: sigma 0.5: mean Sobel 38.9; sigma 1: mean Sobel 31.5; sigma 2: mean Sobel 21.6; sigma 5: mean Sobel 10.3.



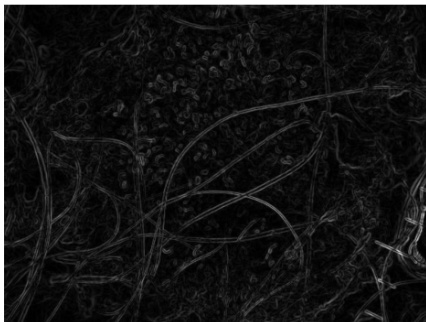
Q3. Main differences between Canny and Sobel

Sobel is one gradient step. It returns a bright ridge everywhere the intensity changes, including weak texture and noise, and the ridge is several pixels wide. Canny starts with a Gaussian blur, then the same Sobel gradients. Non-maximum suppression keeps only the peak of each ridge, so the edge becomes one pixel thick. Hysteresis then keeps strong edges and the weak pixels that touch them, and drops the rest. The output is a binary map, not a brightness image. On ISIC_0051817, the brightest 10% of the Sobel map covers 10.0% of pixels. Canny marks 4.6% of pixels as edge. Canny is thinner and cleaner. Sobel shows the strength of every change.

ISIC_0051817



Sobel magnitude



Canny

